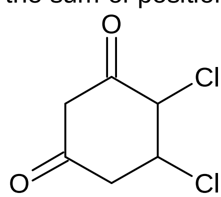
27. For isothermal expansion of an ideal gas into vacuum, among the following, how many are zero?
ΔE, ΔH, ΔT, q, P_{ext}, ΔS_{sys}, ΔS_{surr}, ΔS_{total}, ΔG_{sys}

28. The amount of water produced (in g) in the oxidation of 1 mole of white phosphorus (P_4) with conc. HNO_3 to a compound with highest oxidation state of phosphorus is:

29. Mole fraction of solute in some solution is 1/n. If 50% of solute molecules dissociate into two parts and remaining 50% get dimerised, new mole fraction of solvent becomes 4/5. Find value of n.

30. When the following compound is named correctly according to IUPAC convention, what would be the sum of position of two chlorine atoms?



SECTION – C
(Numerical Answer Type)

*This section contains **TWO (02) paragraphs.** Based on each paragraph, there are **TWO (02)** questions of numerical answer type. The answer to each question is a **NUMERICAL VALUE (XXXXX.XX).** If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.*

Paragraph for Question Nos. 31 and 32

An ideal mixture of liquids A and B with 2 moles of A and 2 moles of B has a total vapour pressure of 1 atm at a certain temperature. Another mixture with 1 mole of A and 3 moles of B has vapour pressure greater than 1 atm. When 4 moles of C are added to 2nd mixture, the vapour pressure comes down to 1 atm. Vapour pressure of C in pure state p_c^0 = 0.8 atm. If vapour pressure of pure A = x atm and vapour pressure of pure B = y atm.

31. The value of x is______?

32. The value of y is______?

Paragraph for Question Nos. 33 and 34

Potassium crystallizes in a body-centred cubic lattice, with an unit cell length $a = 5.20\,\mathring{A}$. If the distance between nearest neighbours is 'x' and the distance between next nearest neighbour is y, then

33. The value of x is_______

34. The value of y is_______